

# Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

## References

- [jqassistant](#)
- [Neo4j Python Driver](#)

## Node Labels

### Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

|    | nodeLabels                                        | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|---------------------------------------------------|---------------------|----------------------------|
| 0  | [Git, Update, Change]                             | 138637              | 46.084832                  |
| 1  | [Git, Change, Create]                             | 44924               | 14.933351                  |
| 2  | [Git, Change, Delete]                             | 22780               | 7.572383                   |
| 3  | [Java, ByteCode, Method, Member]                  | 13477               | 4.479939                   |
| 4  | [Java, ByteCode, Parameter]                       | 13293               | 4.418775                   |
| 5  | [Git, Commit]                                     | 11339               | 3.769238                   |
| 6  | [File, Git]                                       | 8285                | 2.754047                   |
| 7  | [Java, ByteCode, Bound, ParameterizedType]        | 7279                | 2.419639                   |
| 8  | [Java, ByteCode, Bound]                           | 7240                | 2.406675                   |
| 9  | [Git, Change, Rename]                             | 7018                | 2.332879                   |
| 10 | [Java, ByteCode, Member, Field]                   | 3605                | 1.198351                   |
| 11 | [Java, ByteCode, Bound, WildcardType]             | 2947                | 0.979623                   |
| 12 | [Java, Value, ByteCode, Annotation]               | 2931                | 0.974304                   |
| 13 | [Xml, Element]                                    | 2162                | 0.718678                   |
| 14 | [Java, ByteCode, Constructor, Method, Member]     | 2115                | 0.703055                   |
| 15 | [Xml, Text]                                       | 1450                | 0.482000                   |
| 16 | [Java, ByteCode, TypeVariable, Bound]             | 1111                | 0.369312                   |
| 17 | [Java, ByteCode, Method, Member, Lambda]          | 968                 | 0.321776                   |
| 18 | [Type, File, Java, ByteCode, ResolvedDuplicate... | 888                 | 0.295183                   |
| 19 | [Type, File, Java, ByteCode, Class]               | 846                 | 0.281222                   |
| 20 | [Json, Key]                                       | 694                 | 0.230695                   |
| 21 | [Value, Json, Scalar]                             | 678                 | 0.225376                   |
| 22 | [Java, Value, ByteCode, Primitive]                | 672                 | 0.223382                   |
| 23 | [Type, File, Java, ByteCode, JavaType]            | 649                 | 0.215736                   |
| 24 | [Java, ByteCode, Method, Member, GenericDeclar... | 578                 | 0.192135                   |
| 25 | [Type, File, Java, ByteCode, ExternalType]        | 411                 | 0.136622                   |
| 26 | [Git, Change, Copy]                               | 298                 | 0.099059                   |
| 27 | [Author, Git, Person]                             | 295                 | 0.098062                   |
| 28 | [Value, Array]                                    | 287                 | 0.095403                   |
| 29 | [Committer, Git, Person]                          | 242                 | 0.080444                   |

## Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

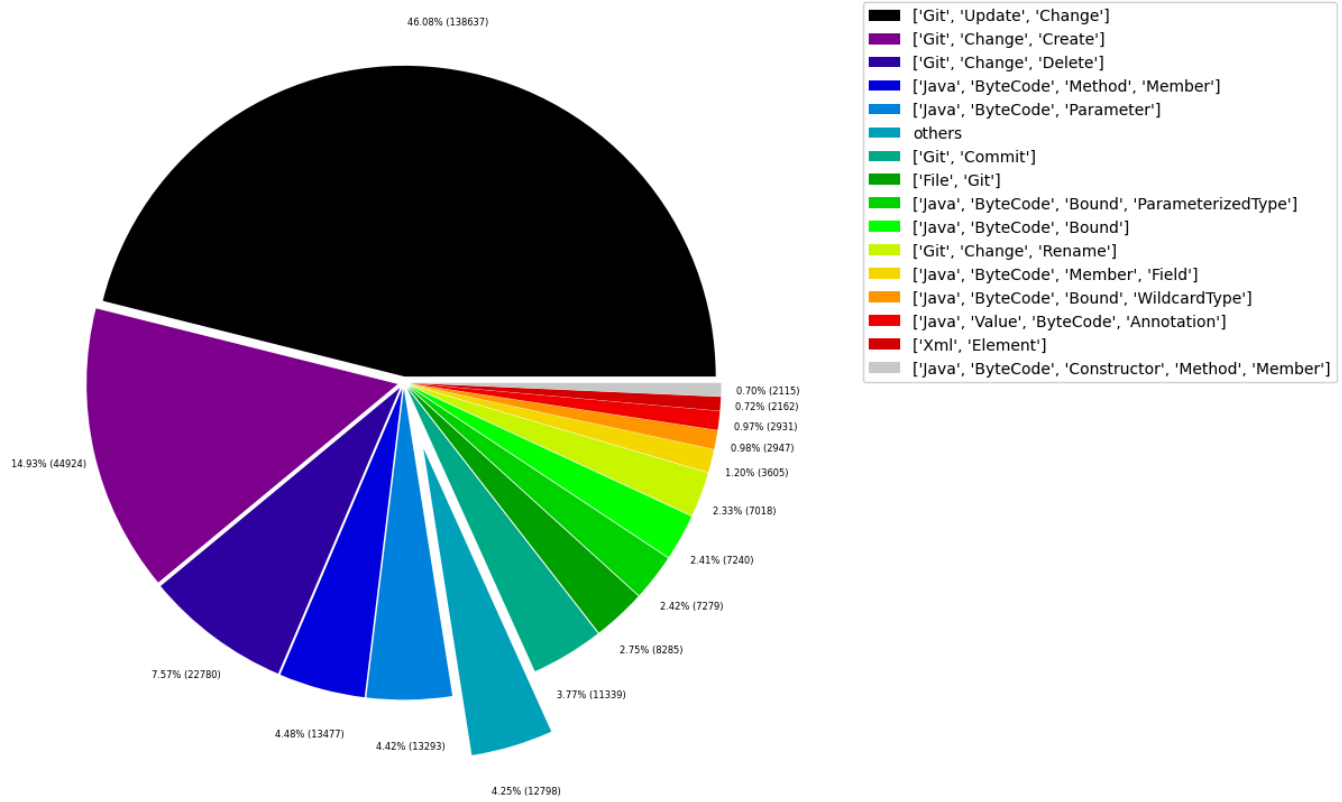


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

|    | nodeLabels                                         | nodesWithThatLabels | nodesWithThatLabelsPercent |
|----|----------------------------------------------------|---------------------|----------------------------|
| 0  | [Analyze, Task, jQAssistant]                       | 1                   | 0.000332                   |
| 1  | [Git, Branch]                                      | 1                   | 0.000332                   |
| 2  | [Repository, File, Git]                            | 1                   | 0.000332                   |
| 3  | [File, Json]                                       | 2                   | 0.000665                   |
| 4  | [File]                                             | 3                   | 0.000997                   |
| 5  | [Java, ByteCode, Constructor, Method, Member, ...] | 4                   | 0.001330                   |
| 6  | [Maven, Exclusion]                                 | 5                   | 0.001662                   |
| 7  | [Value, Array, Json]                               | 6                   | 0.001994                   |
| 8  | [Type, File, Java, ByteCode, Void]                 | 9                   | 0.002992                   |
| 9  | [File, Maven, Xml, Pom, Document]                  | 9                   | 0.002992                   |
| 10 | [Java, ManifestSection]                            | 9                   | 0.002992                   |
| 11 | [File, Artifact, Jar, Archive, Zip, Java]          | 9                   | 0.002992                   |
| 12 | [File, Java, Manifest]                             | 9                   | 0.002992                   |
| 13 | [File, Java, ServiceLoader]                        | 10                  | 0.003324                   |
| 14 | [File, Java, Properties]                           | 12                  | 0.003989                   |
| 15 | [Maven, ExecutionGoal]                             | 16                  | 0.005319                   |
| 16 | [Maven, PluginExecution]                           | 16                  | 0.005319                   |
| 17 | [Xml, Attribute]                                   | 18                  | 0.005983                   |
| 18 | [jQAssistant, Rule, Concept]                       | 19                  | 0.006316                   |
| 19 | [Maven, Plugin]                                    | 21                  | 0.006981                   |
| 20 | [Maven, Configuration]                             | 21                  | 0.006981                   |
| 21 | [Type, File, Java, ByteCode, Throwable, Extern...  | 21                  | 0.006981                   |
| 22 | [Type, File, Java, ByteCode, Throwable, Resolv...  | 22                  | 0.007313                   |
| 23 | [Type, File, Java, ByteCode, Enum]                 | 28                  | 0.009308                   |
| 24 | [Type, File, Java, ByteCode, PrimitiveType]        | 30                  | 0.009972                   |
| 25 | [Xml, Namespace]                                   | 36                  | 0.011967                   |
| 26 | [Type, File, Java, ByteCode, Annotation]           | 44                  | 0.014626                   |
| 27 | [File, Directory]                                  | 44                  | 0.014626                   |
| 28 | [Type, File, Java, ByteCode, Class, Throwable]     | 55                  | 0.018283                   |
| 29 | [Type, File, Java, ByteCode, GenericDeclaratio...  | 85                  | 0.028255                   |

## Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

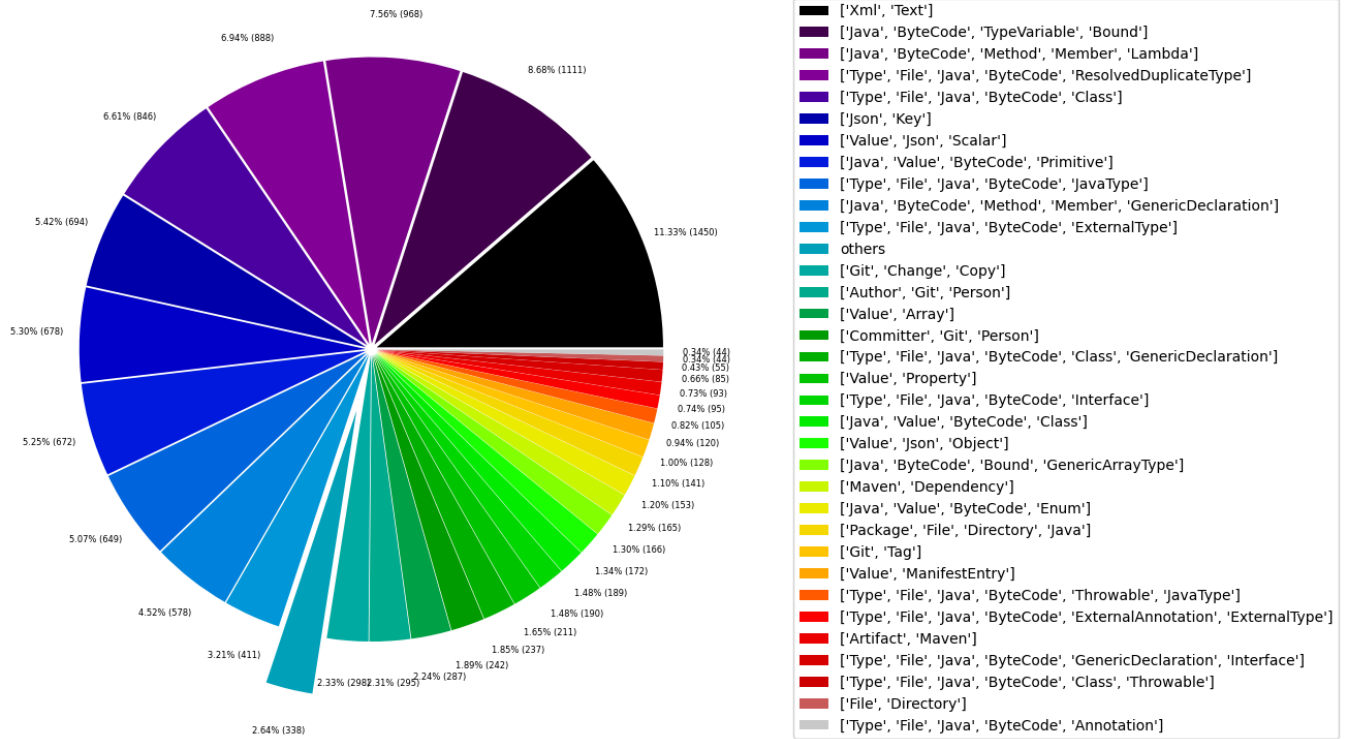


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

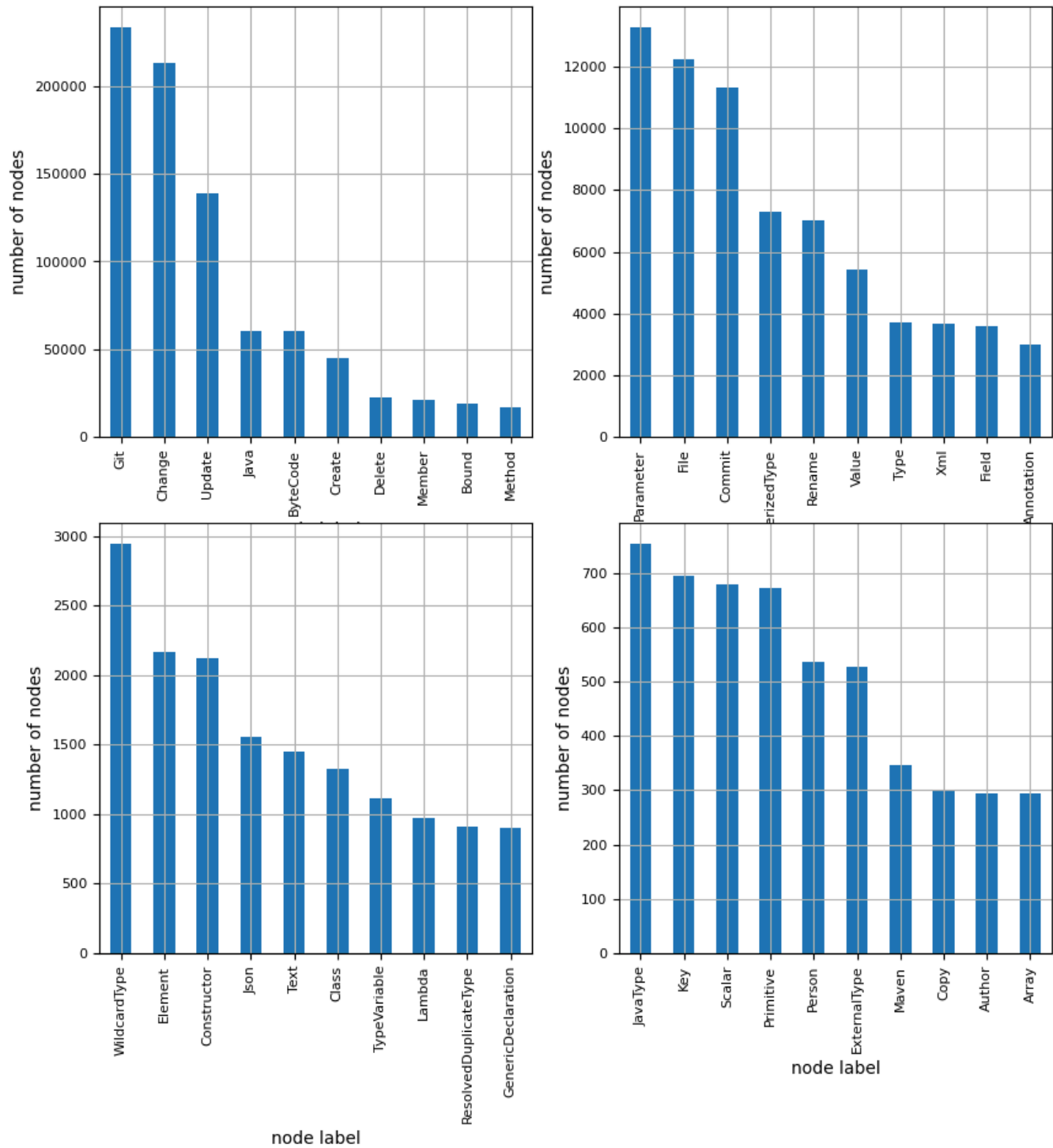
|    | nodeLabel             | nodesWithThatLabel | nodesWithThatLabelPercent |
|----|-----------------------|--------------------|---------------------------|
| 0  | Git                   | 233948             | 77.767510                 |
| 1  | Change                | 213657             | 71.022504                 |
| 2  | Update                | 138637             | 46.084832                 |
| 3  | Java                  | 60633              | 20.155237                 |
| 4  | ByteCode              | 60443              | 20.092079                 |
| 5  | Create                | 44924              | 14.933351                 |
| 6  | Delete                | 22780              | 7.572383                  |
| 7  | Member                | 20747              | 6.896586                  |
| 8  | Bound                 | 18743              | 6.230429                  |
| 9  | Method                | 17142              | 5.698235                  |
| 10 | Parameter             | 13293              | 4.418775                  |
| 11 | File                  | 12240              | 4.068743                  |
| 12 | Commit                | 11339              | 3.769238                  |
| 13 | ParameterizedType     | 7279               | 2.419639                  |
| 14 | Rename                | 7018               | 2.332879                  |
| 15 | Value                 | 5419               | 1.801350                  |
| 16 | Type                  | 3715               | 1.234917                  |
| 17 | Xml                   | 3675               | 1.221620                  |
| 18 | Field                 | 3605               | 1.198351                  |
| 19 | Annotation            | 2975               | 0.988931                  |
| 20 | WildcardType          | 2947               | 0.979623                  |
| 21 | Element               | 2162               | 0.718678                  |
| 22 | Constructor           | 2119               | 0.704385                  |
| 23 | Json                  | 1552               | 0.515906                  |
| 24 | Text                  | 1450               | 0.482000                  |
| 25 | Class                 | 1327               | 0.441113                  |
| 26 | TypeVariable          | 1111               | 0.369312                  |
| 27 | Lambda                | 968                | 0.321776                  |
| 28 | ResolvedDuplicateType | 910                | 0.302496                  |
| 29 | GenericDeclaration    | 904                | 0.300502                  |
| 30 | JavaType              | 754                | 0.250640                  |
| 31 | Key                   | 694                | 0.230695                  |
| 32 | Scalar                | 678                | 0.225376                  |
| 33 | Primitive             | 672                | 0.223382                  |
| 34 | Person                | 537                | 0.178506                  |
| 35 | ExternalType          | 527                | 0.175182                  |
| 36 | Maven                 | 346                | 0.115015                  |
| 37 | Copy                  | 298                | 0.099059                  |
| 38 | Author                | 295                | 0.098062                  |
| 39 | Array                 | 293                | 0.097397                  |

## Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

## Node count by label



## Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 934948

|    | relationshipType         | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|--------------------------|-------------------------------|--------------------------------------|
| 0  | CONTAINS_CHANGE          | 213657                        | 22.852287                            |
| 1  | MODIFIES                 | 213657                        | 22.852287                            |
| 2  | UPDATES                  | 138637                        | 14.828311                            |
| 3  | CREATES                  | 52240                         | 5.587477                             |
| 4  | INVOKES                  | 36558                         | 3.910164                             |
| 5  | DELETES                  | 29798                         | 3.187129                             |
| 6  | COMMITTED                | 22678                         | 2.425589                             |
| 7  | DEPENDS_ON               | 22492                         | 2.405695                             |
| 8  | OF_TYPE                  | 21920                         | 2.344515                             |
| 9  | DECLARES                 | 21208                         | 2.268361                             |
| 10 | OF_RAW_TYPE              | 17353                         | 1.856039                             |
| 11 | HAS                      | 14429                         | 1.543294                             |
| 12 | HAS_PARENT               | 13858                         | 1.482221                             |
| 13 | RETURNS                  | 12844                         | 1.373766                             |
| 14 | HAS_COMMIT               | 11339                         | 1.212795                             |
| 15 | READS                    | 9379                          | 1.003157                             |
| 16 | HAS_ACTUAL_TYPE_ARGUMENT | 8407                          | 0.899194                             |
| 17 | HAS_FILE                 | 8285                          | 0.886146                             |
| 18 | RENAMES                  | 7018                          | 0.750630                             |
| 19 | OF_GENERIC_TYPE          | 5996                          | 0.641319                             |
| 20 | RESOLVES_TO              | 4223                          | 0.451683                             |
| 21 | SIMILAR                  | 4078                          | 0.436174                             |
| 22 | WRITES                   | 3944                          | 0.421842                             |
| 23 | CONTAINS                 | 3926                          | 0.419916                             |
| 24 | HAS_NEW_NAME             | 3662                          | 0.391680                             |
| 25 | RETURNS_GENERIC          | 3593                          | 0.384299                             |
| 26 | CHANGED_TOGETHER_WITH    | 3027                          | 0.323761                             |
| 27 | ANNOTATED_BY             | 2919                          | 0.312210                             |
| 28 | REQUIRES                 | 2230                          | 0.238516                             |
| 29 | HAS_FIRST_CHILD          | 2162                          | 0.231243                             |

## Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>



Relationship types (more than 0.5% overall)

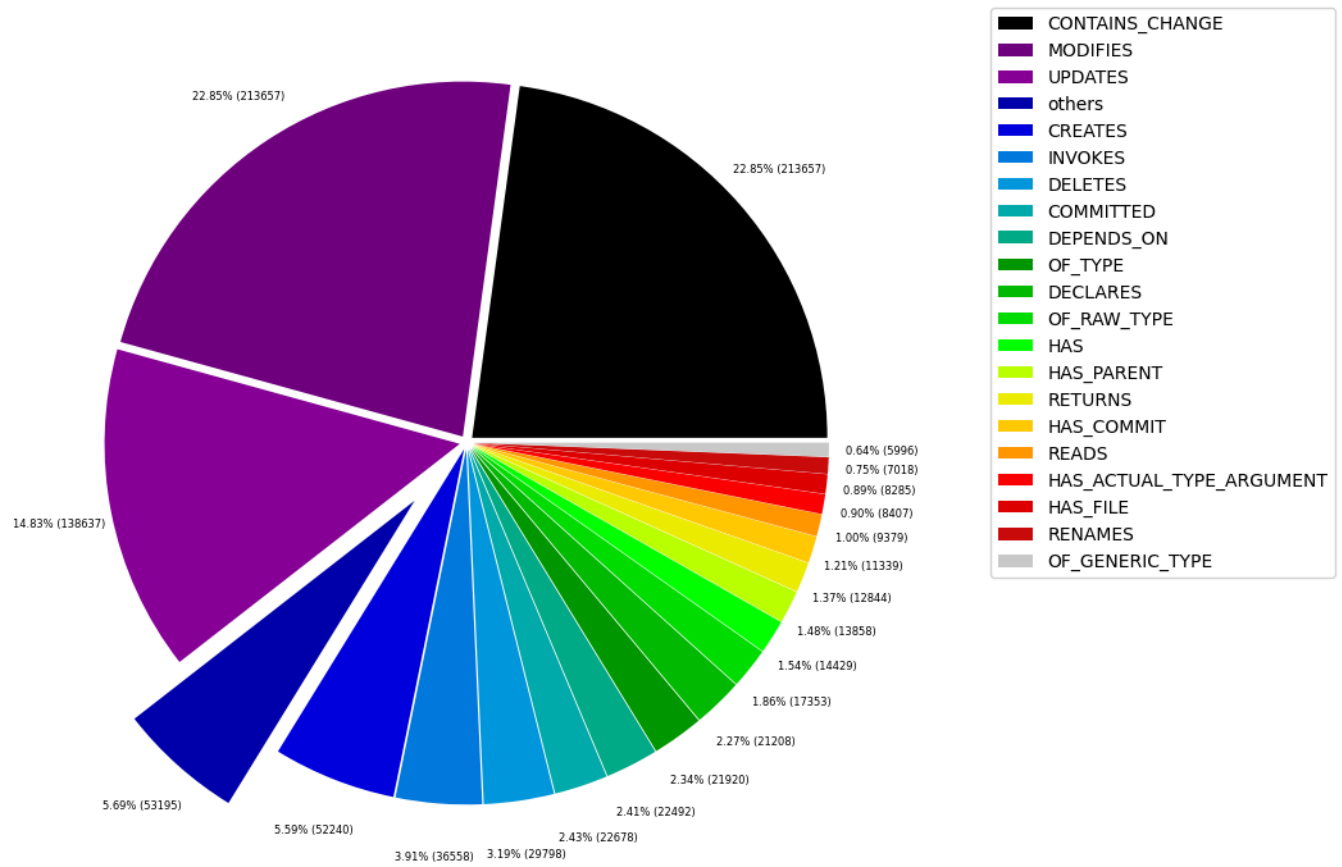


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

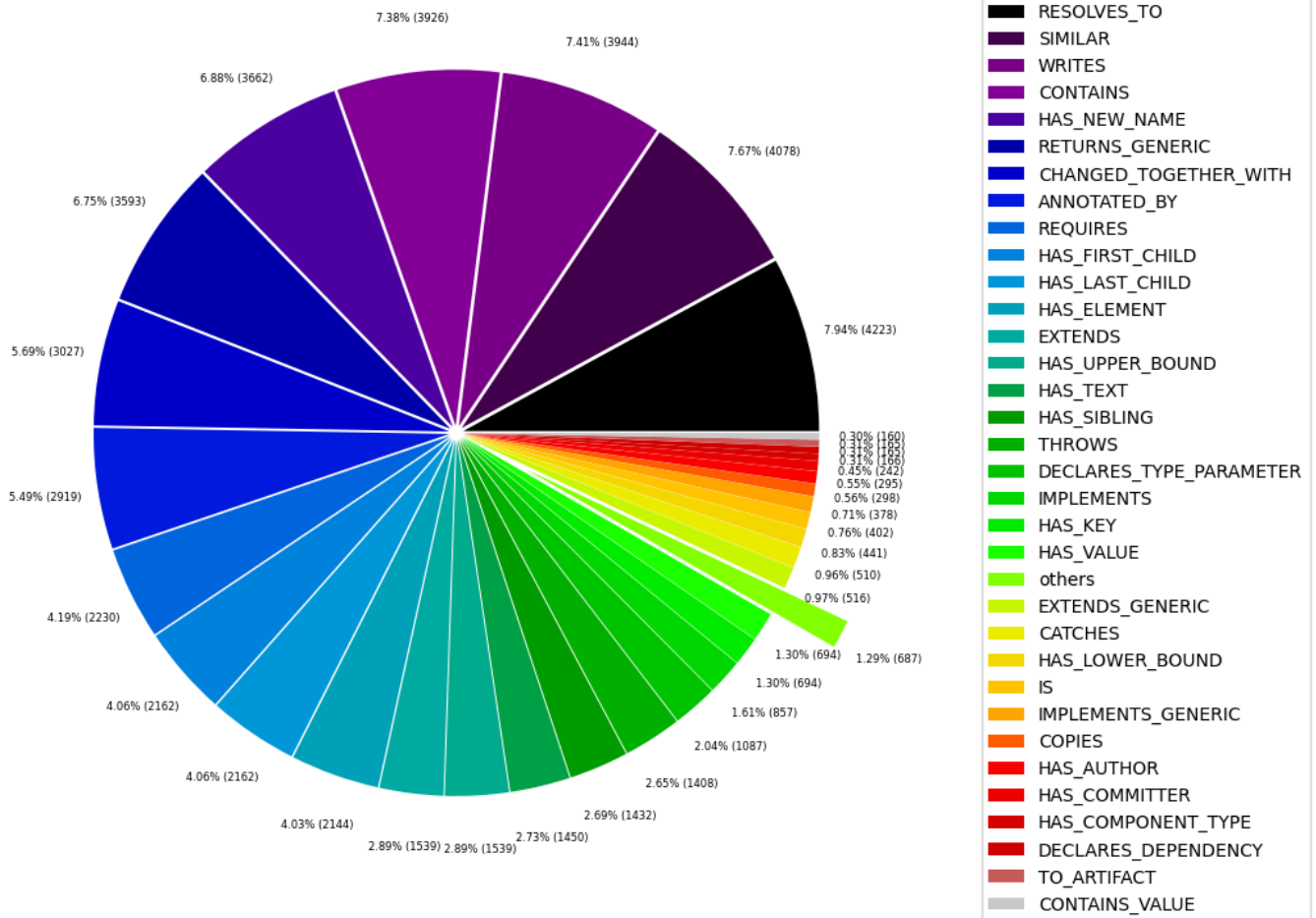
|    | relationshipType        | nodesWithThatRelationshipType | nodesWithThatRelationshipTypePercent |
|----|-------------------------|-------------------------------|--------------------------------------|
| 0  | HAS_PROPERTY            | 1                             | 0.000107                             |
| 1  | HAS_BRANCH              | 1                             | 0.000107                             |
| 2  | HAS_HEAD                | 2                             | 0.000214                             |
| 3  | THROWS_GENERIC          | 5                             | 0.000535                             |
| 4  | EXCLUDES                | 5                             | 0.000535                             |
| 5  | DESCRIBES               | 9                             | 0.000963                             |
| 6  | HAS_ROOT_ELEMENT        | 12                            | 0.001283                             |
| 7  | HAS_GOAL                | 16                            | 0.001711                             |
| 8  | HAS_EXECUTION           | 16                            | 0.001711                             |
| 9  | OF_NAMESPACE            | 18                            | 0.001925                             |
| 10 | HAS_ATTRIBUTE           | 18                            | 0.001925                             |
| 11 | INCLUDES_CONCEPT        | 19                            | 0.002032                             |
| 12 | USES_PLUGIN             | 21                            | 0.002246                             |
| 13 | IS_ARTIFACT             | 21                            | 0.002246                             |
| 14 | HAS_CONFIGURATION       | 21                            | 0.002246                             |
| 15 | REQUIRES_TYPE_PARAMETER | 24                            | 0.002567                             |
| 16 | REQUIRES_CONCEPT        | 28                            | 0.002995                             |
| 17 | DECLARES_NAMESPACE      | 36                            | 0.003850                             |
| 18 | HAS_DEFAULT             | 39                            | 0.004171                             |
| 19 | COPY_OF                 | 119                           | 0.012728                             |
| 20 | HAS_TAG                 | 128                           | 0.013691                             |
| 21 | ON_COMMIT               | 128                           | 0.013691                             |
| 22 | CONTAINS_VALUE          | 160                           | 0.017113                             |
| 23 | TO_ARTIFACT             | 165                           | 0.017648                             |
| 24 | DECLARES_DEPENDENCY     | 165                           | 0.017648                             |
| 25 | HAS_COMPONENT_TYPE      | 166                           | 0.017755                             |
| 26 | HAS_COMMITTER           | 242                           | 0.025884                             |
| 27 | HAS_AUTHOR              | 295                           | 0.031553                             |
| 28 | COPIES                  | 298                           | 0.031873                             |
| 29 | IMPLEMENTS_GENERIC      | 378                           | 0.040430                             |

## Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



## Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

|    | sourceLabels                                  | relationType             | targetLabels                               | numberOfRelationships | numberOfNodesWithSameLabelsAsSource | numbe |
|----|-----------------------------------------------|--------------------------|--------------------------------------------|-----------------------|-------------------------------------|-------|
| 0  | [Git, Update, Change]                         | UPDATES                  | [File, Git]                                | 138637                | 138637                              |       |
| 1  | [Git, Update, Change]                         | MODIFIES                 | [File, Git]                                | 138637                | 138637                              |       |
| 2  | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Update, Change]                      | 138637                | 11339                               |       |
| 3  | [Git, Change, Create]                         | MODIFIES                 | [File, Git]                                | 44924                 | 44924                               |       |
| 4  | [Git, Change, Create]                         | CREATES                  | [File, Git]                                | 44924                 | 44924                               |       |
| 5  | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Change, Create]                      | 44924                 | 11339                               |       |
| 6  | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Change, Delete]                      | 22780                 | 11339                               |       |
| 7  | [Git, Change, Delete]                         | MODIFIES                 | [File, Git]                                | 22780                 | 22780                               |       |
| 8  | [Git, Change, Delete]                         | DELETES                  | [File, Git]                                | 22780                 | 22780                               |       |
| 9  | [Java, ByteCode, Method, Member]              | INVOKES                  | [Java, ByteCode, Method, Member]           | 22243                 | 13377                               |       |
| 10 | [Git, Commit]                                 | HAS_PARENT               | [Git, Commit]                              | 13849                 | 11339                               |       |
| 11 | [Repository, File, Git]                       | HAS_COMMIT               | [Git, Commit]                              | 11339                 | 1                                   |       |
| 12 | [Author, Git, Person]                         | COMMITTED                | [Git, Commit]                              | 11339                 | 295                                 |       |
| 13 | [Committer, Git, Person]                      | COMMITTED                | [Git, Commit]                              | 11339                 | 242                                 |       |
| 14 | [Java, ByteCode, Method, Member]              | HAS                      | [Java, ByteCode, Parameter]                | 8506                  | 13377                               |       |
| 15 | [Java, ByteCode, Method, Member]              | READS                    | [Java, ByteCode, Member, Field]            | 8444                  | 13377                               |       |
| 16 | [Repository, File, Git]                       | HAS_FILE                 | [File, Git]                                | 8285                  | 1                                   |       |
| 17 | [Git, Commit]                                 | CONTAINS_CHANGE          | [Git, Change, Rename]                      | 7018                  | 11339                               |       |
| 18 | [Git, Change, Rename]                         | DELETES                  | [File, Git]                                | 7018                  | 7018                                |       |
| 19 | [Git, Change, Rename]                         | MODIFIES                 | [File, Git]                                | 7018                  | 7018                                |       |
| 20 | [Git, Change, Rename]                         | RENAMES                  | [File, Git]                                | 7018                  | 7018                                |       |
| 21 | [Git, Change, Rename]                         | CREATES                  | [File, Git]                                | 7018                  | 7018                                |       |
| 22 | [Java, ByteCode, Parameter]                   | OF_TYPE                  | [Type, File, Java, ByteCode, JavaType]     | 6156                  | 13293                               |       |
| 23 | [File, Git]                                   | HAS_NEW_NAME             | [File, Git]                                | 3662                  | 8285                                |       |
| 24 | [Java, ByteCode, Bound]                       | OF_RAW_TYPE              | [Type, File, Java, ByteCode, JavaType]     | 3465                  | 7240                                |       |
| 25 | [Java, ByteCode, Bound, ParameterizedType]    | OF_RAW_TYPE              | [Type, File, Java, ByteCode, JavaType]     | 3110                  | 7279                                |       |
| 26 | [Java, ByteCode, Bound, ParameterizedType]    | HAS_ACTUAL_TYPE_ARGUMENT | [Java, ByteCode, Bound, WildcardType]      | 2947                  | 7279                                |       |
| 27 | [Java, ByteCode, Parameter]                   | OF_GENERIC_TYPE          | [Java, ByteCode, Bound, ParameterizedType] | 2695                  | 13293                               |       |
| 28 | [Java, ByteCode, Constructor, Method, Member] | WRITES                   | [Java, ByteCode, Member, Field]            | 2519                  | 2115                                |       |
| 29 | [Java, ByteCode, Bound, ParameterizedType]    | HAS_ACTUAL_TYPE_ARGUMENT | [Java, ByteCode, TypeVariable, Bound]      | 2474                  | 7279                                |       |

## Graph Density

```
total_number_of_nodes (vertices): 300830
total_number_of_relationships (edges): 934948
-> total directed graph density: 1.0331101138252571e-05
-> total directed graph density in percent: 0.0010331101138252572
```